Robed Beauvile-Owen, Ph.D.

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EXPERIENCE

Founder & Data Scientist, *AniPoint Site Research* Pittsburgh, PA | May 2026 - Present

- Founded a car wash site selection consultancy (anipointresearch.com) whose feasibility reports (car-count projections, comparable-site benchmarks) led clients to approve 2 sites with $1.5M+ in projected revenue.
- Built a revenue driver tree (traffic × capture rate × average ticket) and margin models that turned profit goals into car-count targets.
- Estimated that each new car wash adds about $533K to its county's car wash market, using Double ML and causal forests cross-checked with Callaway–Sant'Anna difference-in-differences (within 1.13×).
- Automated Python pipelines for Monte Carlo risk and low/base/high scenarios (230+ unit tests, data-quality checks).

Senior Data Scientist, *Limetree* Dallas, TX (Remote) | September 2024 - April 2026

- Developed uplift and segmentation models on millions of HIPAA-protected records, used in 9 out of 10 campaigns, reducing per-campaign spend by $10K (about 3%) while maintaining baseline enrollment volumes.
- Built an in-house experimentation and reporting platform that automated A/B test design and readouts (power analysis, CUPED variance reduction, significance tests), metric monitoring, and Power BI dashboards, used by the data science and strategy teams to set campaign budgets and saving leadership hours of analysis and readouts each week.
- Added insurance claims and support-ticket complaints to the churn model as leading signals, raising precision by 25% among members with open complaints.
- Translated the annual retention survey analysis from R to Python and automated it end to end, cutting turnaround from 2.5 weeks to 1 day.
- Re-architected the churn pipeline on Spark and Databricks (partitioning, caching, removing repeated joins), cutting runtime by 48% and memory by 23%.

Founding Machine Learning Engineer, *eParts Services LLC* Pittsburgh, PA | September 2023 - September 2024

- Built a regression shipping price model, validated it offline (MAE, human-in-the-loop review) and in A/B tests, and shipped it as a pricing API (C#, Azure, 10,000+ daily requests) that replaced a heuristic rule, cutting the median shipping price error vs. actual carrier cost from $30 to $11 (excluding extreme bulk orders) while maintaining an 18% contribution margin.
- Built automated drift monitoring for live inference pipelines and mentored the software engineering team on production ML practices.
- Cut search complaints by 12% and raised search click-through rate from 30% to 35% by tuning Elasticsearch relevance (custom analyzers, field boosting), validated with NDCG@10, human review, and A/B tests.

Data Scientist, *Limetree* Dallas, TX (Remote) | April 2022 - September 2023

- Raised the offline AUC-ROC of the XGBoost lead-scoring model from 0.62 (prior model) to 0.67 and used SHAP to identify the main drivers (income, age, household composition).
- Confirmed the gain in an A/B test, where model-driven targeting raised gross lead rate (GLR) by 9%.
- Validated GLR out of sample as a 60-day leading indicator of enrollment rate ($R^2 = 0.55$).
- Raised the precision of high-opportunity zone selection by 20% over legacy zoning (scored against past successful campaigns in similar locations), using spatial overlays and demographic indexing in GeoPandas.
- Led monthly monitoring of up to 10 production models built for clients (accuracy, feature drift, forecast vs. actual) and built a company-wide self-service dashboard on curated Snowflake and Databricks tables.

TECHNICAL PROJECTS

Medicare Claims Intelligence & Fraud Detection medicare-claims-intelligence

- Built a dbt warehouse (4 staging models, 5 marts, 26 data tests) over 790K CMS claims, with tests that caught 11,253 claims missing a service date.
- Built an out-of-time XGBoost risk model (SHAP, MLflow) that found 46% more next-year high-cost members than a prior-year-cost baseline in the same top-5% list, beat a seasonal-naive forecast in a 35-origin rolling backtest (MASE 0.62 vs. 1.06), and validated fraud flags on injected test anomalies (81% precision at review capacity).

EDUCATION

Ph.D. in Mathematics, *University of Pittsburgh* 2017 - 2023

Dissertation: A Minimal Invariant Set, Frames, and Operators

B.S. in Mathematics, *Florida State University* 2014 - 2017

Minor: Philosophy | **GPA:** 3.90

TECHNICAL SKILLS

Metrics & Planning: KPI Frameworks, KPI / Driver Trees, Standardized Metric Definitions, Leading Indicators, Acquisition & Retention Marketing Analytics, Contribution Margin, Scenario Analysis, Target Setting, Forecasting

Causal Inference & Experimentation: A/B Testing, Experimentation Platforms, CUPED, Power Analysis, Double ML, Causal Forests, Difference-in-Differences, Uplift Modeling

Modeling & Statistics: Regression, Classification, XGBoost, Time Series Forecasting, Monte Carlo Simulation, SHAP, Drift Monitoring

Data & Tools: SQL (Snowflake, SQL Server), Python (pandas, GeoPandas, statsmodels, scikit-learn, econml, MLflow), R (survey analysis), C#, Databricks, Spark, dbt, Elasticsearch, Azure, Power BI, Excel, Git